Preston Yoshino

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Education

Grinnell College Grinnell, IA
Bachelor of Arts in Computer Science, Mathematics *Aug. 2023 – May 2027*

- Relevant Coursework: Analysis of Algorithms, Abstract Algebra, Linear Algebra, Differential Equations, Statistical Modeling, Graph Theory, Data Science, Statistical Modeling, Machine Learning, Real Analysis, Probability

Experience

Firmware Developer Intern May 2026 – Present
International Business Machines *Poughkeepsie, NY*

- Designed and implemented an agentic workflow using LangGraph to autonomously analyze performance data on IBM z/OS through multi-step reasoning pipelines
- Built MCP tools to enable seamless interaction with enterprise data stores, streamlining data access and retrieval across distributed systems
- Developed CI/CD pipelines and automated testing suites to accelerate deployment cycles and ensure code reliability across production environments

Artificial Intelligence Researcher Sep. 2025 – Present
University of Iowa *Iowa City, IA*

- Formalized adaptive temperature scheduling in Group Relative Policy Optimization as a stochastic control problem, deriving dynamic learning rate adjustments from empirical reward standard deviation to optimize exploration-exploitation tradeoffs
- Modeled group reward variance within the GRPO objective function to construct a temperature-aware policy gradient variant, targeting reduction of convergence steps in RL-based language model training through principled statistical regularization

Software Engineer Oct. 2025 – Mar. 2026
John Deere *Waterloo, IA*

- Designed and developed internal-facing web applications using TypeScript, React/Next.js, and SQL, improving workflow efficiency in over 200 factories across the United States and Mexico
- Containerized and orchestrated microservices with Docker and Kubernetes, integrating CI/CD pipelines to streamline deployments and ensure scalable, cloud-ready environments

Projects

SoilSense | *PyTorch, Torchvision, React, Python, FastAPI, Docker* Jan. 2026 – May 2026

- Formulated precision agriculture as a probabilistic multi-class classification problem over geospatial feature vectors, fine-tuning two EfficientNet-B0 PyTorch classifiers achieving 95% validation accuracy
- Fused CNN outputs with satellite, elevation, and weather API data streams to construct 15×15 crop-suitability probability grids upsampled via interpolation to 150×150 contour maps
- Orchestrated VLM inference and Llama-3.2-3B crop recommendations through a Dockerized FastAPI backend

PrestonGPT | *PyTorch, HuggingFace Transformers, PEFT/LoRA, TRL, Mistral-7B, FastAPI* June 2026 – Present

- Fine-tuned dense and classification layers of a Mistral-7B model via LoRA on personal text data to encode individual writing style and tone
- Built a reward model from human-labeled preference pairs and applied RLHF (PPO) to align outputs with personality-specific targets

Technical Skills

Languages: Python, JavaScript, Java, C, C++, SQL, Rust, PyTorch, CUDA, TensorFlow, Keras
Frameworks: React, Node.js, Flask, Databricks, AWS, FastAPI, PostgresSQL, MongoDB, Express.js